TARRIK QUNEIBI

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EXPERIENCE

ISS Corporate (formerly SustainaBase) - *Senior Associate Sustainability Data Engineer* Feb 2023 - Present

- **Architected and deployed cloud-native AWS Infrastructure-as-Code pipelines** to deliver consistent, audit-ready emissions disclosures following emissions reporting standards. Reducing processing cycles from weekly turnarounds to seconds, accelerating client reporting and reducing internal overhead.
- **Designed and managed multi-tiered database architecture** integrating relational, NoSQL, and data lakehouse systems to support scalability, large-scale data storage, versioning, enrichment, and rapid querying. Enabling both internal and external analytics across multiple greenhouse gas emission categories.
- **Provided environmental consulting to clients** by leveraging expertise in GHG emissions accounting, regulatory reporting frameworks, and data engineering to support emissions tracking goals and resolve complex technical questions. This improved reporting accuracy, ensured audit readiness across emissions disclosure requirements, and accelerated progress toward emissions reduction targets.

University of Michigan - *Research Assistant / Project Lead* Jan 2022 - Feb 2023

- **Managed city-wide water quality research** evaluating UV disinfection's effect on opportunistic pathogens in drinking water distribution systems. Coordinated scheduling, field sampling logistics, and task delegation across municipal water treatment facility personnel, local businesses, and university research staff.
- **Conducted comprehensive water quality assessments** including physicochemical characterization, biological sampling, and plant operational data collection across multiple distribution system locations.
- **Established reproducible data management workflows** integrating field measurements, laboratory analysis, and treatment plant operational data into standardized formats with API-driven pipelines, enabling consistent reporting and versioning across the project team.

City of Ann Arbor Drinking Water Treatment Plant - *Water Resource Engineering Intern* May 2021 - Jan 2022

- **Executed an instrumentation validation study** for newly installed post-filtration particle counters. Developed controlled experiments using calibrated particle suspensions to verify detection reliability for Cryptosporidium-sized oocysts.
- **Built a time-series data model** to analyze continuous particle count streams. Applying statistical methods to correlate filtration performance indicators (turbidity, flow rate, and runtime) with pathogen (surrogate) particle breakthrough events during daily flow rate variations.
- **Developed a decision-support dashboard** by integrating the data model with disinfection contact time parameters, enabling plant operators to make informed, data-driven treatment adjustments.

SKILLS

Languages: Python, Java, JavaScript, R

Databases & Storage: SQL Server, MySQL, DynamoDB, MongoDB, S3 Data Lake Architecture

Cloud & Infrastructure: AWS Cloud Infrastructure (EC2, S3, RDS, Lambda, Redshift, Stepfunctions)

Modeling & Reporting: Carbon Accounting & Reporting (GHG Protocol, CSRD, CDP), Geospatial analysis (arcGIS), Hydraulic/Hydrologic modeling (Modflow, SWMM, EPANet), Process modeling (SUMO), Data Visualization (Custom built, Sisense, Tableau)

Machine Learning: Neural Networks, Classification, Gradient Boosted Decision Tree, Linear Regression

EDUCATION

Master of Science in Environmental Engineering - The University of Michigan Dec 2022

Bachelor of Science in Environmental Engineering - The University of Michigan Dec 2021